Last update: August 2023

Somatoform disorders: diagnosis and treatment

Author: Dr. Sibel Gezer-Dickschat

Fachärztin für Gynäkologie und Geburtshilfe

Co-Author(s): Prof. Dr. med. Brigitte Leeners, Dr. med. Judith Alder, Prof. Dr. med. Johannes Bitzer

Definition and classification

Physical symptoms that cannot (or cannot sufficiently) be attributed to a physical disease. The severity of the symptoms is such that everyday function is disrupted.

Monosymptomatic form

- One persistent symptom

Polysymptomatic form

- Several physical symptoms insufficiently explained by physical disorders at the same time and over time, sometimes years

Examples

- Chronic pelvic pain
- Vulvodynia
- Urogynecological diseases (overactive bladder, chronic recurrent cystitis)
- Mastodynia

How can doctors identify patients with somatoform disorders?

- Incoherence between subjective suffering and objective findings
- Patient believes in a physical cause and persistently requests medical investigations in spite of negative results
- Multiple somatic investigations
- Patient resistant to external referral to mental health specialists
- Difficult doctor-patient relationship: Hope-disappointment cycle (see below)
- Frequent change of doctor (doctor shopping, doctor hopping, doctor swapping)
- Thick, often chaotic medical file
- Feelings of powerlessness and disappointment in patient and medical professionals

Classification of somatoform disorders according to ICD-10 (F45)

F 45.0 Somatization disorder

- Rare extreme form, multiple symptoms of at least two years duration

F 45.1 Undifferentiated somatoform disorder

- Common less extreme variant, less striking

F 45.2 Hypochondriacal disorder

- Anxious preoccupation with cause dominates

F 45.3 Somatoform autonomic dysfunction

- Symptoms involving organs under autonomic innervation
- F45.30 Cardiovascular system
- F45.31 Upper gastrointestinal tract
- F45.32 Lower gastrointestinal tract
- F45.33 Respiratory system
- F45.34 Genitourinary system
- F45.38 Other organs and systems
- F45.39 Unspecified organs and systems

F 45.4 Persistent somatoform pain disorder

- Predominant complaint is pain
- F 45.41 Chronic pain disorder with somatic and psychological factors

F 45.8/9 Other/unspecified somatoform disorders

Single-symptom classification

For individual functional syndromes, a single-symptom classification is possible, for example:

- K 58: Irritable bowel syndrome
- K 30: Functional dyspepsia
- M 79.70: Fibromyalgia
- G 93.3: Postviral fatigue syndrome
- N 94: Pain and other conditions associated with female genital organs and menstrual cycle

Classification according to ICD-11: Disorders of bodily distress or bodily experience

Bodily Distress Disorder (no symptom-based differentiation). Additional identification according to severity:

- Mild Bodily Distress Disorder

- Moderate Bodily Distress Disorder
- Severe Bodily Distress Disorder
- Body Integrity Dysphoria

Classification according to Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-V) (published June 2013): Somatic Symptom Disorder (SSD)

Replaces: Somatization disorder, somatoform disorders, hypochondriasis, pain disorder

Criteria A, B (at least 1) and C must be met:

A) One or more somatic symptoms that are distressing or result in significant disruption of daily life.

B) Excessive thoughts, feelings, or behaviors related to the somatic symptoms or associated health concerns as manifested by at least one of the following:
1. Disproportionate and persistent thoughts about the seriousness of one's symptoms.
2. Persistently high level of anxiety about health or symptoms.
3. Excessive time and energy devoted to these symptoms or health concerns.

C) Although any one somatic symptom may not be continuously present, the state of being symptomatic is persistent (typically more than 6 months).

* Specify if: With predominant pain (previously: pain disorder): This specifier is for individuals whose somatic symptoms predominantly involve pain.

In the ICD-11 and DSM-V, the corresponding disorders are classified as Bodily Distress Disorder (ICD-11) and Somatic Symptom Disorder. They are no longer listed under neurotic disorders and subtypes are omitted.
The key difference with somatoform disorders is that somatic symptom disorder (DSM-V) and bodily distress disorder (ICD-11) can also be diagnosed in case of symptoms of physical origin if psychological features dominate.

The most important differential diagnoses are:

- Other physical conditions
- Panic disorder

- Generalized anxiety disorder
- Depressive disorders
- Illness anxiety disorder
- Dissociative disorders (disorder with functional neurological symptoms)
- Delusional disorder
- Body dysmorphic disorder
- Obsessive-compulsive disorder

Epidemiology

In gynecology, 66 % of symptoms are medically unexplained (often not yet meeting the criteria for a somatic disorder) [P01]. The majority of cases improve/regress within two weeks.

- Prevalence of people with multiple somatic symptoms (at least 4 symptoms in men and 6 in women): 5-11 %, 12-month prevalence of somatoform disorders in women (Germany): 15% [P02].
- Higher proportion in outpatient care: 17–40 %
- Women are significantly more affected: Somatization disorder 5–10:1, Pain disorder 2:1, Hypochondriasis 1:1
  [P01][P02]

Comorbidity with other mental disorders, particularly depression and anxiety, is high.

Etiology

Iatrogenic factors affecting the emergence and continuation of somatoform disorders

- Overlooking the patient's emotional cues
- Repeated somatic diagnosis for reassurance
- Sharing suspected physical diagnoses prematurely (e.g. before completing investigations)
- Incorrect interpretation of trivial/incidental findings e.g. on MRI
- Prescription of medication despite uncertain diagnosis
- Only negative explanation of symptoms ("There's nothing wrong")
- Lack of central coordination of individual investigations and communication with other HCPs
- Lack of clarification of biological, psychological and social factors

Specific factors affecting the emergence and continuation of somatoform disorders

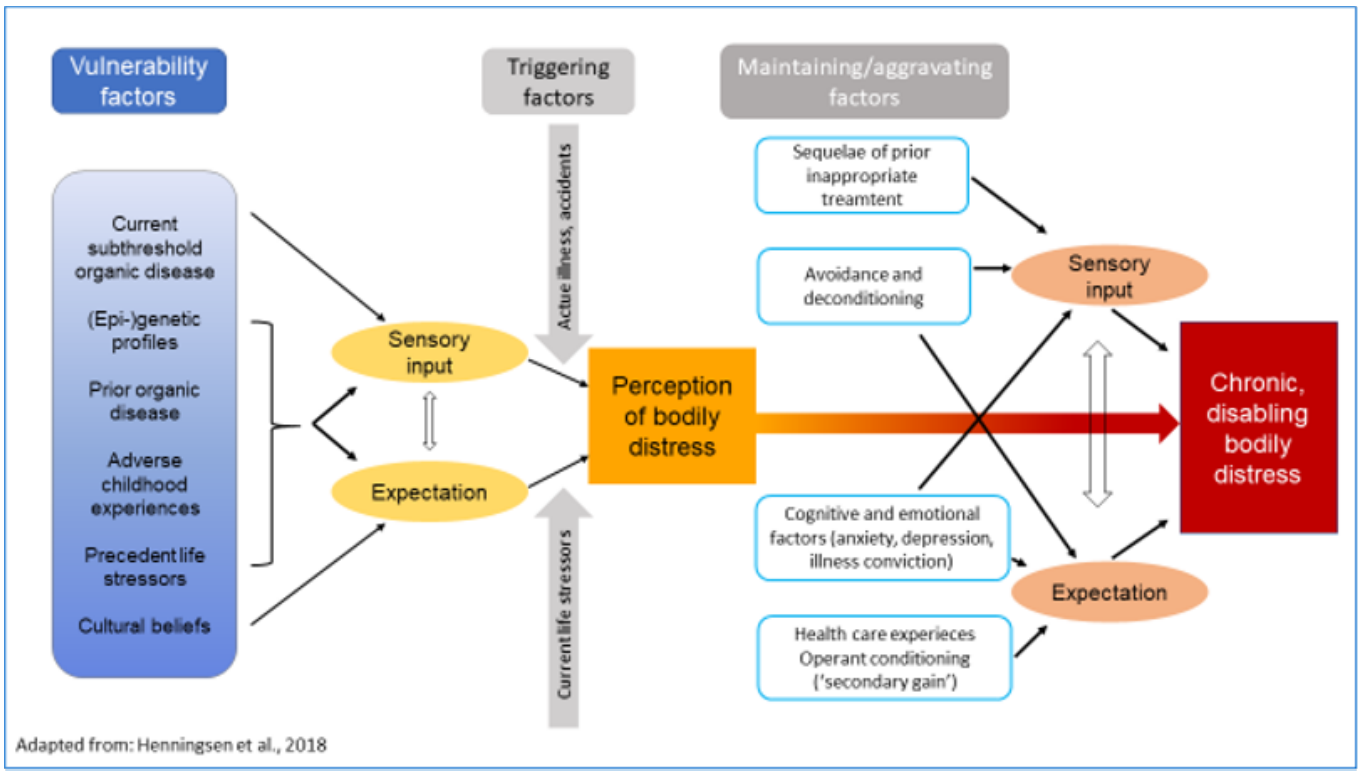

Figure 1: Disorder model of functional physical symptoms (with kind permission of Prof. Dr. med. Peter Henningsen).

Pathophysiology of somatoform disorders

Somatoform disorders have a multifactorial origin with a probable predominance of psychosocial factors and their somatic correlates.
In a simplified view:
Long-term stress → Chronic activation of stress response centers with impact on immunological processes → Internal organ dysfunction, tissue damage due to inflammatory processes, for example → Chronic nociceptor inflammation → Reduced pain threshold, central sensitization associated with adrenal insufficiency at the endocrinal level → Disassociation of sensation of pain from original inflammatory trigger and independence of pain

Psychophysiological model of somatoform disorders (learning theory)

In addition to these suggested, and in some cases confirmed, pathophysiological mechanisms, the psychophysiological model represents the patient's specific reaction modes in an understandable way:

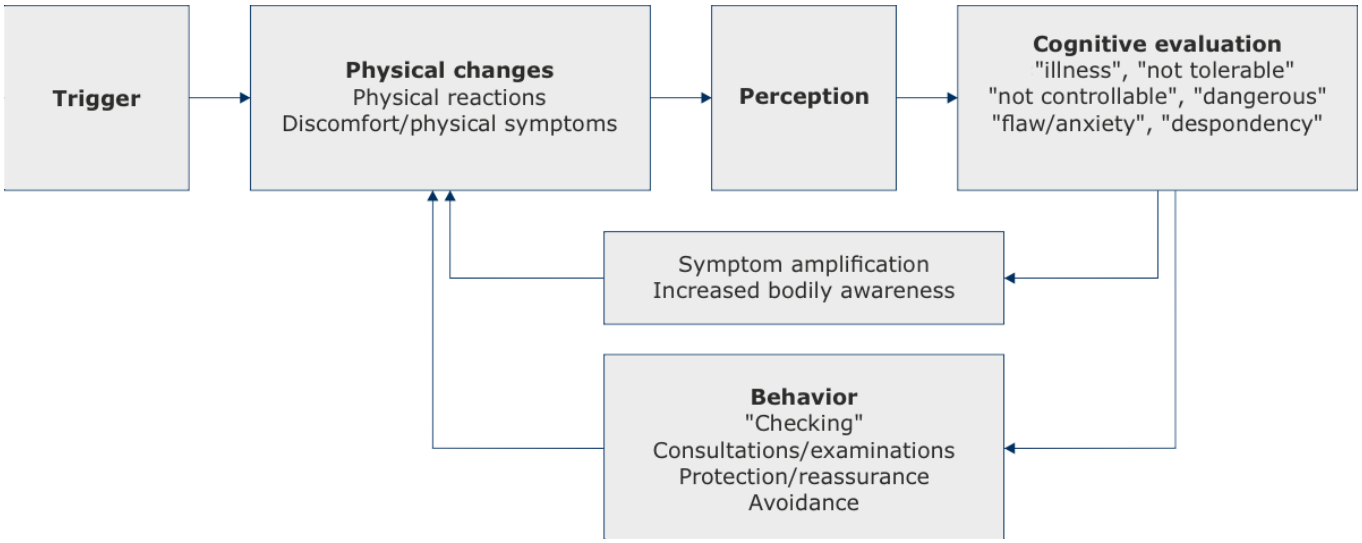

Figure 2: Psychophysiological model of somatoform disorders.

Key assumptions for managing patients

- Patients with somatoform disorders are no more, but no less, physically ill than other patients. For this reason, physical conditions must be ruled out repeatedly over time, for example if symptoms change suddenly.
- The symptom is the disease. The symptom is the result of an impaired relationship between biological, psychological and social factors.
- Symptoms are real (patients feel pain and are not “pretending”) and should not be questioned or psychologized. The cause of the symptoms is physical, not mental, but they have ‘taken on a life of their own’ and lead to psychological stress.
- The question is less “why?” (cause) and more “how” the symptoms have emerged (analysis of circumstances and behavior).
- Somatoform disorders present non-fatal chronic disruptions to quality of life; the focus is on care, not cure.
- Somatization disorders have far-reaching social effects, which are often gender-specific.

Treatment objectives

- Early identification
- Secondary prevention (avoidance of iatrogenic fixation and chronic disorder)
- Extension of the patient’s explanatory model from a preoccupation with purely physical causes to a wider biological, psychological and social understanding of the condition.
- Patient feels accepted, respected and supported; reduced anxiety.
- The patient can establish their own connections between their situation (“stress”), thoughts, feelings and physical symptoms and can develop an understanding of how the body, mind and environment interact.
- They learn ways to influence their body through mental training and their mind through physical training.
- They develop an understanding of quality of life and an idea of how to control their quality of life through changes in their behavior.
- They learn about themselves and their life story. They gain insight and new social skills. Their personal growth is encouraged.
- There may be increased motivation to undergo psychotherapy.

Diagnosis

The diagnosis of somatoform disorders always includes two dimensions:

Physical diagnosis (by exclusion)

Sufficient physical diagnosis of the physical symptoms presented is necessary. However, repeated, unnecessary physical diagnostics for exclusionary purposes bear the risk of the patient further fixating on a purely physical explanation of their condition. A physical diagnosis by exclusion alone is not a sufficient basis for diagnosis.

> Planned, non-redundant diagnostic investigations carried out over a short period, coordinated by a “case manager” who consults with other doctors.

Psychological diagnosis

Comorbidity with other psychological disorders, particularly depression and anxiety, is high. The psychological diagnosis must therefore go beyond the quantification of symptoms and take into account current emotions, conflictual and stressful situations, personal difficulties and social and cultural factors.

A diagnostic recording of psychological factors should be carried out on the first indication of a somatoform disorder, not on completion of all necessary physical investigations (avoiding the notion of “all in their head”).

- Over 50 % of patients with unexplained physical symptoms reveal psychosocial factors spontaneously.

> If a somatoform disorder is suspected, expand the history beyond the leading symptoms: other current symptoms, previous course of symptoms and treatment, indications of other psychological impairments, current psychosocial difficulties and trigger situations, factors perpetuating the disorder, personal background.

Questionnaire for Evaluating Somatoform Disorders

The PHQ-15 is a module of the Patient Health Questionnaire (PHQ) and is used to establish the severity of somatic symptoms. The total “somatic symptoms” scale covers 15 somatic symptoms, corresponding to the 15 most common physical symptoms in outpatients and the most important DSM-IV criteria for somatization disorder. As it has been shown that the presence or absence of a physical cause has no relevance for the diagnosis of somatoform disorder, the criterion of a physical explanation is omitted from the diagnosis of somatoform disorder in DSM-V.

The element of the PHQ-15 that requires no physical explanation for the physical symptoms can therefore be used to measure the severity of somatic symptoms and the severity of somatization.

Patient Health Questionnaire-15 (PHQ-15)

Clinical management

Downward spiral of an unhelpful patient-doctor relationship

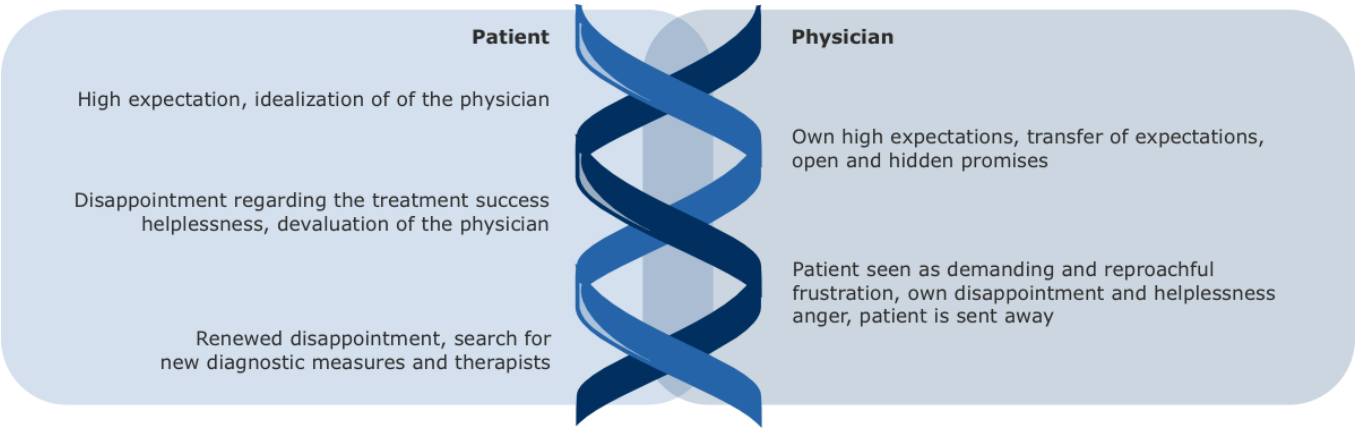

Figure 3: Downward spiral of an unhelpful patient-doctor relationship.

Preventing the downward spiral

- Self-awareness for the doctor: What feelings does the patient trigger in me?
- Take the patient's complaints seriously
- Do not take on the patient's pressure: "lean back", composed attitude, treatment objective is care not cure
- Clear setting (time structure, regular appointments, etc.)
- Patient information: share diagnosis, plan further investigations, share models

Phase 1: Active listening, encouraging narrative (G17A)

Objective	Patient feels respected and understood, doctor receives all the information needed to build a picture of the patient's reality
Knowledge and skills	Active listening, establishing a therapeutic relationship, giving the patient time and space to tell their story, partnership, empathy, respect, coping-oriented
Tasks	Description of disorder in personal and situational context Which symptoms, for how long, with what effects on QoL, symptom management
Helpful tools/ interview techniques	Record information (e.g. show evolution of symptoms on a time chart), use Visual Analogue Scale to evaluate symptoms

Phase 2: Biological, psychological and social history and assessment (G17A)

Objective	Collection of relevant background information: History, previous treatment, expectations for treatment, descriptive classification of symptoms and understanding of the patient's illness model
Knowledge and skills	Somatic and psychosocial history taking, understanding the biopsychosocial model, asking about meanings and emotions, switching between open and closed questions, checking back; summarizing
Tasks	Simultaneous recording of biological and psychosocial data, initial assignment for descriptive classification, understanding Presenting layperson hypotheses
Helpful tools/ interview techniques	"What other physical symptoms have you had recently?" "How do you feel about your symptoms?" "What do YOU think could be connected to the symptoms?" Ask about exceptions: "In the last 24 hours, when did you feel worse than you do right now? What was happening?" Homework: Symptom diary including description of the circumstances, cognitive-emotional evaluation process and behavior vis-a-vis symptoms

Phase 3: Exploring causes and developing a collaborative disorder model

Objective	Develop one or more hypotheses for causal factors for the disorder; derive and arrange further investigations
Knowledge and skills	Knowledge of somatic diseases and pathologies, knowledge of psychosocial factors affecting health, didactic and explanatory skills
Tasks	Collect information on maintenance/situational analysis and add to an initial cause model, taking into consideration layperson hypotheses, suggesting and initiating appropriate further investigations
Helpful tools/ interview techniques	Present summarized information in graph form

Phase 4: Explanatory diagnosis

Objective	Patient and doctor have a shared understanding of the disorder and determining factors
Knowledge and skills	Knowledge of biopsychosocial model of health and disease with biological, psychological and social causes at different times (predisposing, precipitating, maintaining)
Tasks	Develop an understanding of the problem/symptom taking into consideration biological, psychological and social factors with etiological importance, summarizing and discussing the determining factors for the specific patient
Helpful tools/ interview techniques	9 Factor Matrix

Factor matrix	Biological	Psychological	Social
Predisposing	Family risks, pregnancy and birth related risks	Early trauma, abuse, neglect	Broken family, early separation, migration
Precipitating	Disease, drugs, biological transition	Loss, life transition, separation	Migration, cultural norms, social changes
Maintaining	Side effects of drugs etc.	Anxiety, false beliefs, stress responses	Secondary reinforcement in the environment

Figure 4: The 9 Factor Matrix of Biopsychosocial Diagnosis

Phase 5: Defining treatment objectives

Objective	Collaborative determination of objectives for change at various levels, e.g. symptoms, work, prevention
Knowledge and skills	Open questions are support for individual responsibility through confirmation, identifying and encouraging the use of resources, brainstorming, sharing information
Tasks	Identify and prioritize objectives on several levels, agree objectives with patient, clarify realistic and less realistic objectives, include an element of flexibility

Helpful tools/ interview techniques	Define clear success criteria: How do doctor and patient identify objectives that have been reached

Phase 6: Finding solutions, developing a treatment plan

Objective	Patients gain new insights into possible connections and learn about new possible solutions. They then decide on one or more paths independently.
Knowledge and skills	“How can the patient influence their body through their thoughts and behaviors?”, knowledge of treatment options and sharing of this knowledge: Psychophysiological treatments, self-assessments, perception and awareness training, relaxation techniques, behavioral training: exposure, control behaviors and protective behaviors, communication training, QoL improvements e.g. sport.
Tasks	Establish an individual treatment plan taking into account the patient’s expectations and possibilities, usually as a multidisciplinary collaboration with somatic and psychosocial interventions
Helpful tools/ interview techniques	See below (Psychotherapeutic interventions)

Phase 7: Joint evaluation and support

Objective	Offer the patient check-ups and reassurance in case of questions or anxiety; further development of the concept of health, ongoing adaptation to the learning and healing process.
Knowledge and skills	Knowledge of processes of change and evaluation criteria, verification and adaptation of hypotheses, knowledge of networking opportunities, case management, frustration tolerance and patience
Tasks	Determine and verify evaluation criteria. Record treatment successes and failures Implement adjustments, new decisions, referrals as applicable or introduction requests

Helpful tools/ interview techniques	

Psychotherapeutic treatment approaches

Criteria for referral to psychotherapy

A psychotherapy referral, discussed in advance with the patient, should take place if:

- No improvement is observed after six months of appropriate psychosomatic treatment or the severity/acuity of symptoms deems this appropriate
- The patient is unable to work for more than four weeks at a time
- There is a suspicion of additional acute mental disorders or obvious personal factors with a strong influence
- The patient desires psychotherapeutic treatment
- Interactional issues with the patient complicate treatment

Despite the limited evidence, psychotherapy should be clearly recommended due to the findings. Most evidence is available for the effect of cognitive behavioral therapy (CBT).

Psychotherapeutic interventions

Learning physical controls on “neutral” territory

- Relaxation techniques
- Biofeedback
- Body scan meditation

Behavioral experiments

- Directing attention to change bodily perception

Provoking physical reactions

- Stair-climbing
- Abandoning protective behaviors
- Objective: Develop a neutral concept of health and increase tolerance of (normal) physical reactions

Encouraging positive experiences and behaviors

- Encourage positive experiences and management
- “Enjoyment training”
- Increased activity (physical, social, cultural, work, leisure)
- “Distraction”
- Encourage social interactions/skills

Pharmacotherapy

In cases of comorbid depression or anxiety disorders, drug-based treatment and/or psychotherapeutic treatment should be considered.
Temporary use of analgesics may relieve complaints and symptoms.

Literature

Nr.	Title	Author	Journal	Volume	Pages
[P01]	Medically unexplained symptoms: an epidemiological study in seven specialities	Nimmnuan C, Hotopf, M, Wessley, S	J Psychosom Res	51(1)	361-367
[P02]	DSM-IV pain disorder in the general population. An exploration of the structure and threshold of medically unexplained pain symptoms	Fröhlich C, Jacobi F, Wittchen HU	Eur Arch Psychiatry Clin Neurosci	256(3)	187-196
[P03]	The Patient Health Questionnaire Somatic, Anxiety, and Depressive Symptom Scales: a systematic review	Kroenke K, Spitzer RL, Williams JB, Löwe B	Gen Hosp Psychiatry	32(4)	345-359
[P04]	Vulvodynia as a possible somatization disorder. More than just an opinion	Mascherpa F, Bogliatto F, Lynch PJ, Micheletti L, Benedetto C	J Reprod Med	52(2)	107-110
[P05]	The 'evil twin syndrome' in chronic pelvic pain: A systematic review of prevalence studies of bladder pain syndrome and endometriosis	Tirlapur SA, Kuhrt K, Chaliha C, Ball E, Meads C, Khan KS	Int J Surg		

Nr.	Title	Author	Journal	Volume	Pages
[P06]	Systematic review of therapies for noncyclic chronic pelvic pain in women	Yunker A, Sathe NA, Reynolds WS, Likis FE, Andrews J	Obstet Gynecol Surv	67(7)	417-425
[P07]	Eisenberg Center for Clinical Decisions and Communications Science. Treating Chronic Pelvic Pain: A Review of the Research for Women. Comparative Effectiveness Review Summary Guides for Consumers	John M	Agency for Healthcare Research and Quality		
[P08]	Behavioural and psychological management of somatic symptom disorders: An overview	Sharma MP, Manjula M	Int Rev Psychiatry	25(1)	116-124
[P09]	Qualitative research as the basis for a biopsychosocial approach to women with chronic pelvic pain	Souza PP, Romão AS, Rosa-e-Silva JC, Reis FC, Nogueira AA, Poli-Neto OB	J Psychosom Obstet Gynaecol	32(4)	165-172
[P10]	ACOG Practice Bulletin No. 51. Chronic pelvic pain	ACOG	Obstet Gynecol	103(3)	589-605
[P11]	Chronic pelvic pain	Howard FM	Obstet Gynecol	101(3)	594-611

Nr.	Title	Author	Journal	Volume	Pages
[P12]	WHO systematic review of prevalence of chronic pelvic pain: a neglected reproductive health morbidity	Latthe P, Latthe M, Say L, Gulmezoglu M, Khan KS	BMC Public Health	6	177
[P13]	Chronic pelvic pain: prevalence, health-related quality of life, and economic correlates	Mathias SD, Kuppermann M, Liberman RF, Lipschutz RC, Steege JF	Obstet Gynecol	87(3):	321-327
[P14]	A randomized clinical trial to compare two different approaches in women with chronic pelvic pain	Peters AA, van Dorst E, Jellis B, van Zuuren E, Hermans J, Trimbos JB	Obstet Gynecol	77(5)	740-744
[P15]	Musculoskeletal causes of chronic pelvic pain: a systematic review of existing therapies: part II	Tu FF, As-Sanie S, Steege JF	Obstet Gynecol Surv	60(7)	474-483
[P16]	Screening for musculoskeletal causes of pelvic pain	Prendergast SA, Weiss JM	Clin Obstet Gynecol	46(4)	773-782
[P17]	Definition, Pathophysiologie, Diagnostik und Therapie des Fibromyalgiesyndroms	AWMF 041/004			

Nr.	Title	Author	Journal	Volume	Pages
[P18]	The prevalence of endometriosis in women with chronic pelvic pain	Guo SW, Wang Y	Gynecol Obstet Invest	62(3)	121-130
[P19]	Patterns of diagnosis and referral in women consulting for chronic pelvic pain in UK primary care	Zondervan KT, Yudkin PL, Vessey MP, Dawes MG, Barlow DH, Kennedy SH	Br J Obstet Gynaecol	106(11)	1156-1161
[P20]	The role of laparoscopy as a diagnostic tool in chronic pelvic pain	Howard FM	Baillieres Best Pract Res Clin Obstet Gynaecol	14(3)	467-494
[P21]	Consensus statement for the management of chronic pelvic pain and endometriosis: proceedings of an expert-panel consensus process	Gambone JC, Mittman BS, Munro MG, Scialli AR, Winkel CA	Fertil Steril	78(5)	961-972
[P22]	Prevalence of Chronic Pelvic Pain Among Women: An Updated Review	Ahangari, A	Pain Physician	17(2)	
[P23]	Approach to the patient with multiple somatic symptoms	Croicu, C., Chwastiak, L., & Katon, W	Med Clin North Am	98(5)	1079-1095

Nr.	Title	Author	Journal	Volume	Pages
[P24]	Current advancements in the diagnosis and treatment of chronic pelvic pain	Morrissey, D., Ginzburg, N., & Whitmore, K	Curr Opin Urol	24(4)	336-344
[P25]	Clinical value of DSM IV and DSM 5 criteria for diagnosing the most prevalent somatoform disorders in patients with medically unexplained physical symptoms (MUPS)	van Dessel, N. C., van der Wouden, J. C., Dekker, J., & van der Horst, H. E	J Psychosom Res	82	4-10
[P26]	Intensivkurs Psychiatrie und Psychotherapie	Lieb K, Frauenknecht S.		9. Auflage	
[P27]	AWMF S3 Leitlinie «Funktionelle Körperbeschwerden» (Reg-Nr. 051-001l)	AWMF			